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DRAFT Program Solicitation
Fleetwood
BIOLOGICAL TECHNOLOGIES OFFICE

NOTE – this is a DRAFT version of the Fleetwood Program Solicitation. To that end, the information listed below (and throughout) is subject to change before the final opportunity is posted.

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PROGRAM SOLICITATION OVERVIEW

- **Federal Agency Name** – Defense Advanced Research Projects Agency (DARPA), Biological Technologies Office (BTO)
- **Funding Opportunity Title** – Fleetwood
- **Announcement Type** – DRAFT
- **Dates/Time - All Times are Eastern Time Zone (ET)**
 - Industry Day: March 12, 2026
 - Submission Deadlines TBD

Description of the Funding Opportunity: The Defense Advanced Research Projects Agency (DARPA) is soliciting proposals to develop catalysts to enable atomic-level precision and control of complex systems at all scales of production. Fleetwood is seeking pathways to significantly improve the ability to valorize lignin-containing waste biomass feedstocks, both domestically and in deployed environments, by converting them into resources and precursors for value-added chemicals primed for disruption. Catalyst development focuses on identifying pathways to convert lignin into chemicals that are traditionally imported into the U.S. Efforts also focus on improving catalyst stability, maximizing activity and selectivity, and ensuring the catalysts are feedstock agnostic, capable of functioning regardless of biomass sourcing.

- **Multiple awards are anticipated.**
- **Types of instruments that may be awarded** – Other Transaction for Prototype (OT-P) Agreements
- **Agency Contact**

The Solicitation Coordinator for this effort may be reached at:
Fleetwood@darpa.mil
DARPA/ BTO
675 North Randolph Street
Arlington, VA 22203-2114
- **Attachments**

Attachment A: Abstract Summary Slide Template
Attachment B: Abstract Template

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PROGRAM SOLICITATION
Defense Advanced Research Projects Agency (DARPA)
Fleetwood

1. PROGRAM INFORMATION

1.1 Background

The goal of the Fleetwood program is to develop stable, active, and selective catalysts that are capable of valorizing lignin-containing waste biomass feedstocks, both domestically and in deployed environments, thereby accessing a robust source for value-added chemicals and precursors.

Current supply chain logistics rely on complex production systems and transportation networks to provide petroleum-derived chemicals for commercial and military applications. Increasingly, manufacturers depend on foreign-sourced imports to produce finished products, posing significant economic and strategic risks to U.S. supply chains. The conversion of waste streams, such as municipal, agricultural, and logging waste, has been explored as a potential route to produce molecules that can augment supplies of traditional petrochemicals. Industry struggles to utilize abundant lignocellulosic feedstocks to produce value-added chemicals as lignocellulosic feedstocks are notoriously difficult to process and refine. Lignin, a key component, is nature's largest repository of aromatic compounds but is often treated as a waste product. Current conversion methods like gasification, pyrolysis, and hydrothermal liquefaction suffer from low yields, poor selectivity, and high energy requirements, limiting their economic viability. The Fleetwood program seeks to overcome these limitations by developing transformative catalytic technologies that enable atomic-level precision and control, unlocking the full potential of lignin as a valuable resource.

The program objective is accomplished through two Focus Areas (FAs). Abstracts and OPPs must address both FAs in each Phase to be considered for evaluation. The first focuses on developing approaches to extract lignin from a variety of feedstocks, and the second focuses on ensuring catalyst stability while maximizing activity and selectivity to convert lignin into chemicals traditionally derived or sourced from imports. These approaches will be evaluated using comprehensive technoeconomic analysis (TEA) and process modelling across three production scales to assess economic viability and scalability.

Teaming is highly encouraged to ensure that proposers possess the expertise and capabilities necessary to meet Fleetwood's objectives.

1.2 Acquisition Strategy

The abstract submission deadline is anticipated to be April 6, 2026, 5:00PM (Eastern Time). Abstracts will be reviewed by the Government; if selected, the proposer will be asked to brief an Oral Presentation Package (OPP). OPPs will be reviewed by the Government, and if selected, they may result in a Phase 1 award of an Other Transaction (OT) for Prototype and eligibility to participate in future Phases of the program.

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This Program Solicitation (PS) encourages solutions from all responsible sources capable of satisfying the Government's needs, including large and small businesses, nontraditional defense contractors as defined in 10 U.S.C. § 3014, and research institutions as defined in 15 U.S.C. § 638.

To facilitate this objective, the Government will use the following acquisition process for Fleetwood:

1.2.1 Abstracts

Through this solicitation, the Government requests proposers to submit abstracts (see Section 4.2) in response to both Phases and FAs. The Government will review all submitted abstracts for technical comprehension and ability (see Section 4.3). Selected proposers will be invited to brief OPPs in an Oral Presentation (see Section 4.4) to the Government.

1.2.2 Oral Presentations

Upon the Government's request, proposers will have the opportunity to present their OPP to the Government program team. Additional details and guidance for preparing the OPP will accompany the invitation to participate. The Government will evaluate all OPPs (see Section 4.5) from teams whose abstracts were selected. Oral presentations of the OPP are required and will afford the Government an opportunity to ask clarifying questions of the briefing teams. The OPP, including the briefing, will require significantly more detail than the abstracts, and will provide detailed information needed for Phase 1 selection decisions. The Government anticipates that teams selected based on the OPP and OPP briefing will be awarded a Phase 1 Other Transaction (OT) for Prototype award to address Phase 1 goals/metrics over a 12-month period of performance with a 12-month Option for Phase 2 work.

1.3 Program Scope

1.3.1 Overall Scope of the Fleetwood Program

The Fleetwood program will explore the efficient and effective utilization of lignin, a complex and underutilized resource derived from wood and plant biomass, to create resilient supply chains and enable point of need manufacturing of chemicals. Lignin is nature's largest repository of aromatic compounds, which are valuable and historically difficult to synthesize. The Fleetwood Program will create advanced catalyst designs and demonstrate lignin valorization and conversion into value-added chemicals. The program objective is accomplished through two Focus Areas (FAs). The first focuses on developing approaches to extract lignin from a variety of feedstocks, and the second focuses on ensuring catalyst stability while maximizing activity and selectivity to convert lignin into chemicals traditionally derived or sourced from imports.

1.3.2 Focus Area 1 (FA1): Novel Catalytic Pathways for Accessing Lignin from Real Biomass

FA1 focuses on developing catalytic pathways to extract lignin from real biomass sources (feedstocks) and demonstrating effective separation and conversion of lignin into monomers.

Abstracts and OPPs must include catalysts and processes that are capable of feedstock-agnostic operation by tuning process parameters (including but not limited to temperature, pressure, flow rate, etc.).

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Biomass feedstocks could include, but are not limited to: rice residue, banana residue, coconut residue, coffee residue, nut shells/hulls, palm fronds, corn stover, sugar cane bagasse, soybean residue, macadamia nut shells/hulls, sargassum seaweed, switchgrass, elephant grass, forest residues, oil palm residues, wheat residues, and barley residues.

Performers should propose an approach suitable to multiple types of biomass feedstocks (e.g. woody biomass and agricultural residue, or agricultural residue and grasses). However, the Program Manager will have the discretion to incorporate a replacement biomass feedstock(s) for each performer.

Achieving high delignification rates in a feedstock-agnostic process affords a direct route to increase carbon conversion efficiency for whole-biomass from under 30% to near or above 80%. This is accomplished by making the carbon fraction in lignin accessible for catalytic processing, unlocking its potential for further conversion into value-added chemicals.

While numerous depolymerization strategies effectively cleave aryl-ether C-O bonds, few methods address the catalytic cleavage of C-C bonds in the remaining aromatic oligomers. Developing methods to cleave both C-O and C-C bonds effectively allows for 100% monomer yields which unlocks numerous downstream chemicals and potential applications. Example approaches may include, but are not limited to, coupling reductive depolymerization with subsequent oxidative depolymerization to convert remaining lignin dimers and oligomers into monophenolics.

1.3.3 Focus Area 2 (FA2): Catalyst Development for the Conversion of Lignin to Value-added Chemicals

FA2 focuses on developing catalytic pathways to selectively convert delignification products extracted in FA1 into value-added chemicals. Catalytic approaches are not limited to a single type; thermo-, electro-, photo-, and cell-free biocatalysis are all considered within scope and may be combined in hybrid approaches. Teams are encouraged to propose multi-component catalytic systems that integrate multiple types of catalysis and catalysts to achieve the desired molecular targets. Abstracts and OPPs must detail the current state of the art for a chosen catalytic pathway and a path to achieve industrially relevant catalyst properties in the life of the program. Abstracts and OPPs must define state of the art and target improvements in these areas during the life of the program.

Molecules of interest include, but are not limited to pharmaceuticals, defense-relevant materials, lubricants, and other specialty and value-added chemicals. The design of catalytic systems must prioritize selectivity and activity, ensuring the formation of target chemicals at high yield while minimizing side products. Fleetwood seeks approaches that maximize carbon utilization and enhance the overall economic viability of the biomass conversion process.

Abstracts and OPPs should include routes to functional substitutes or performance advantage chemicals versus direct product replacements. Proposers should evaluate atom economy and energy usage in addition to cost when selecting a chemical(s). Additionally, abstracts and OPPs must specify and justify the production of the target chemical across three scales of production, providing rationale for scalability and feasibility at each level.

1.3.4 Program Constraints/Out of Scope

Abstracts and OPPs must address both FAs in each Phase to be considered for evaluation. Failure to do so will result in the proposal being deemed non-conforming and excluded from

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further consideration. Addressing both FAs ensures proposed approaches provide a complete route for converting whole biomass to chemicals and adequately attend to challenges associated with multi-step catalytic biomass separation and conversion.

Abstracts and OPPs must focus on novel catalytic strategies that align with the program's emphasis on advancing state-of-the-art technologies for atomic-level precision and control in lignin valorization and conversion, while ensuring the economic viability of the proposed approaches through efficient resource utilization and cost-effective implementation.

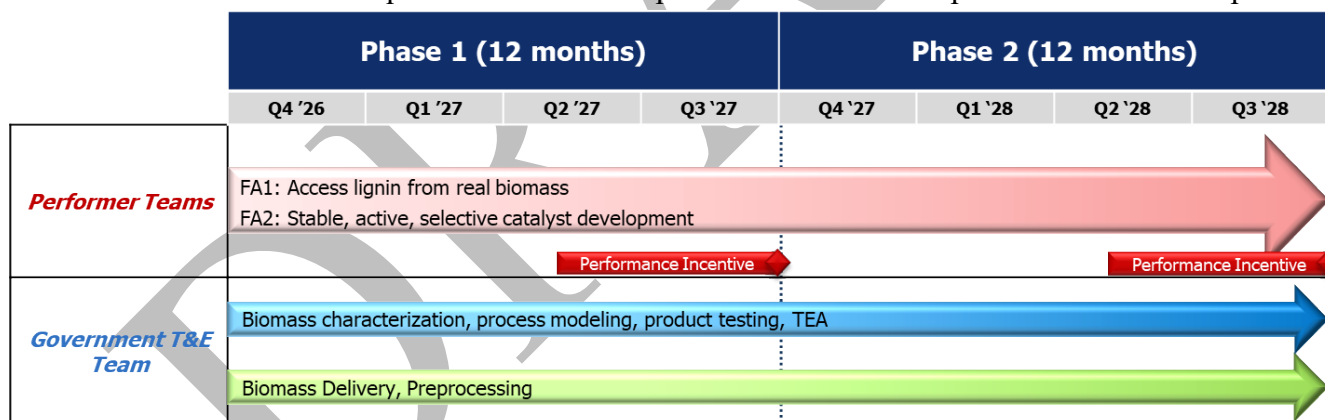
Performers will develop catalytic strategies to produce value-added chemicals across three production scales and should focus on maximizing overall value as assessed through a comprehensive technoeconomic analysis (TEA) and process model. Overall value will be determined through an analysis of the Minimum Selling Price of the chosen product chemicals. Strategies to only produce low-value chemicals, including C1 and C2 products, such as carbon dioxide, methanol, and ethanol are not of interest; however, these chemicals may be reaction intermediates or secondary reaction products for chosen approaches.

The following approaches are deemed out of scope:

Abstracts and OPPs including pyrolysis, gasification, or hydrothermal liquefaction will be deemed non-conforming.

1.4 Program Structure

The Fleetwood program is a 24-month total effort with a 12-month Phase 1 and a 12-month Phase 2. Performer teams are required to address and price both FAs in each phase to ensure a compliant



submission.

Focus Area 1 (FA1):

Performers will develop and test catalytic systems to extract and convert lignin from biomass feedstocks, such as but not limited to rice residue, palm fronds, coconut residue, and banana residues. Biomass feedstocks will be sourced and supplied by the Government Test and Evaluation (T&E) Team to ensure consistency across the program. Success will be measured by:

- The percentage of delignification achieved from the input feedstock
- Demonstrating effective separation and conversion of lignin into monomers

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Approaches must demonstrate stability during extended operation, high selectivity, and near complete depolymerization of lignin into monomers, which will serve as precursors for downstream conversion into value-added chemicals.

Focus Area 2 (FA2):

Performers will develop catalysts to convert lignin-derived monomers into value-added chemicals including but not limited to specialty chemicals, pharmaceuticals, adhesives, and lubricants. Successful strategies must demonstrate high selectivity, activity, yield, efficiency, and stability. C1 and C2 chemicals are not considered value-added chemicals for this effort and should not be targets for proposed approaches. Abstracts and OPPs should address the following over three production scales:

- Catalyst process parameters and tunability
- Process stability and catalyst regeneration
- Process output recycling and maximizing process intensification
- Reduction of required unit operations and improvements in economic metrics

Success will be measured by:

- Maximizing space-time yield (STY, $\text{kg}_{\text{product}}/\text{day}/\text{L}_{\text{reactor}}$)
- Demonstrating stability, activity, and selectivity of the catalytic approach

Techno-Economic Analysis (TEA)

In Phase 1, performers will develop and provide a comprehensive TEA for their solution. In Phase 2, this analysis is updated based on Phase 1 progress and results.

This assessment must 1) evaluate technical feasibility, including technology readiness level (TRL), integration risks, and projected development milestones and 2) analyze economic viability, including cost structure, anticipated return on investment (ROI), pricing strategy, long-term stability, and potential for scalability.

Moreover, the TEA should include market determination and sizing, as well as clearly outlining assumptions, methodologies, and data sources. The analysis should also include relevant regulatory and reimbursement pathways, timelines, cost implications, and demonstrate:

- Alignment with market needs
- Economic justification for adoption and deployment
- Strategic insight into commercialization pathways

Abstracts and OPPs must propose three production scales for chosen value-added chemicals ranging from small-scale, modular unit production (e.g. forward deployed manufacturing) to large-scale chemical production. During the program, performers will evaluate their TEA at each of these production scales, which may be updated and reevaluated during the period of performance. Performers are encouraged to account for biomass resource availability when planning and evaluating production scales as biomass transportation, storage, and preprocessing costs are often overlooked and represent a significant portion of overall cost for any effort. Evaluation of multiple production scales will identify both chemicals and associated scales that are commercially viable

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while exploring the potential for multiple, modular production sites to reduce capital and operating expenditures. Performers are encouraged to include modeling of routes to valorize carbohydrate fractions as well as lignin fractions. These chemicals can be recycled in the lignin conversion process or directly as a product which may be useful to maximize the value of whole biomass.

Performance Incentive

Subject to availability of funds, one performer may be awarded a performance incentive of \$50,000 at the conclusion of each 12-month phase based on Minimum Selling Price (MSP) of the chosen output chemical(s) and demonstrating maximum overall value of the process. This evaluation will account for capital expenditure (CAPEX), operating expense (OPEX), maintenance costs, and other economic considerations, through verification and validation of the performers' TEA and MSP by the Program Manager and the Government T&E Team.

Program Meetings and Locations:

Anticipated program meetings and locations including performer site visits and demonstrations are found below in Table 1.

Meeting Type	Anticipated Location	Frequency
Phase Kickoff	Arlington, VA	Once per phase
Phase Demonstration	Performer Site	Once per phase (Month 11 and Month 23)
Site Visit	Performer Site	Up to 3x per phase
Principle Investigator (PI) Meeting	Arlington, VA	Once per phase
Technical & Programmatic Updates	Tele/videoconference	At least monthly

Table 1: Fleetwood program meetings and locations.

In order to ensure that appropriate progress is being made towards meeting the prescribed metrics, performer teams will be required to demonstrate their catalytic systems – or, when applicable, specific subsystems – to the Government T&E Team and Program Manager, at pre-defined times during the program phases. Each demonstration will have a specific purpose related to the particular goals of the given phase.

The program will include two critical demonstrations, scheduled for Month 11 and Month 23, hosted by the performer teams at their respective performance sites. These demonstrations are aligned with milestones and deliverables outlined in Table 3 (Section 1.6) and are designed to validate progress and ensure that performer teams remain on track to meet program objectives.

1.5 Program Goals/Metrics

Success in the Fleetwood program will be defined by meeting or exceeding the metrics outlined below in Table 2, which represent the minimum set of requirements established by the Government.

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Key Metrics: The program metrics focus on the design and demonstration of catalytic systems capable of converting real biomass into value-added chemicals. These metrics are feedstock-agnostic and apply broadly to all performer approaches. Proposing teams are strongly encouraged to include additional quantitative metrics tailored to their specific designs and strategies to further characterize success. The Government reserves the right to add team-specific metrics to awards as necessary to manage risks associated with individual approaches.

Calculation of lignin depolymerization, or lignin yield, and aromatic monomer yield must follow standard, accepted research practices such as those set forth in Abu-Omar et al.¹ Characterization of biomass samples should be performed in accordance with US Department of Energy National Laboratory of the Rockies – Laboratory Analytical Procedures (LAPs) for relevant characterization techniques. Catalyst characterization and reactor studies should be performed following accepted research practices such as those set forth in the above citation.

OPPs will include routes to functional substitutes or performance advantage chemicals versus direct product replacements. Proposers are encouraged to evaluate atom economy and energy usage in addition to cost when selecting a chosen chemical(s).

Overall Program Metrics			
Category	Rationale	Phase 1	Phase 2
Delignification	Evaluate and optimize flexibility of biomass processing with varied feedstocks – lignin yield on a mass basis	≥50%	≥90%
Monomer Yield	Demonstrate efficient separation and conversion of lignin to monomers – aromatic monomer yield on a mass basis	≥50%	100%
Process Rate	Efficient conversion needed to improve yield and inform sizing – STY (space-time yield)	Maximize Space-Time Yield	
Catalyst Development	Demonstrate stability, activity, and selectivity of developed catalysts	Catalysts maintain performance across biomass feedstocks (stable, active, selective)	
Performance Incentive			
Category	Rationale	Phase 1	Phase 2
Performance Incentive	Maximize overall value of products as represented by MSP, accounting for CAPEX, OPEX, maintenance, and	Evaluated based on Minimum Selling Price (MSP)	

¹ “Guidelines for performing lignin-first biorefining” Energy Environ. Sci., 2021, 14, 262

	other costs – verified by Government T&E Team	
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Table 2: Fleetwood program metrics and milestones.

1.6 Program Deliverables

In addition to the metrics above, performers are expected to provide monthly and quarterly status updates to the Government, including technical and financial summary reports. The Fleetwood program is structured around major technical milestones distributed across Phase 1 and Phase 2. Each milestone corresponds to a critical area of advancement in the field of biomass valorization and catalytic system design. Performers are expected to achieve, at minimum, the below milestones with associated deliverables:

Milestones and Deliverables for all Focus Areas	
Phase 1:	1 Month – <u>Milestone:</u> Complete program kickoff meeting. <u>Deliverable:</u> Kickoff slide deck summarizing research goals and technical plan. <u>Deliverable:</u> Preliminary System Design Report. 3 Months – <u>Milestone:</u> Develop initial TEA process model for chosen catalyst system for selected biomass types. Develop target catalysts following initial biomass studies (e.g., forestry residue, agricultural residue, etc.). <u>Deliverable:</u> Report summarizing milestone data. 6 Months – <u>Milestone:</u> Demonstrate delignification, monomer yield, process rate, and catalyst stability improvement for multiple types of whole biomass (four or more sources, e.g., corn stover, palm frond, coconut husk). <u>Deliverable:</u> Report summarizing milestone data. 11 Months – <u>Milestone:</u> Demonstrate delignification, monomer yield, process rate, and stability improvement consistent with End of Phase Metrics. <u>Deliverable:</u> Report summarizing milestone data. <u>Deliverable:</u> Provide TEA and process model for T&E evaluation. End of Phase Milestone – 12 Months: <u>Milestone:</u> Achieve $\geq 50\%$ delignification and $\geq 50\%$ monomer yield with improved Space Time Yield; T&E will evaluate chemical value with comprehensive TEA/process model. <u>Deliverable:</u> End of Phase report summarizing progress against all tasks, metrics, and milestones.
Phase 2:	15 Months – <u>Milestone:</u> Refine TEA process model for chosen catalyst system for selected biomass types. Optimize target catalysts on whole biomass sources (e.g., forestry residue, agricultural residue, etc.). <u>Deliverable:</u> Report summarizing milestone data. 18 Months – <u>Milestone:</u> Demonstrate delignification, monomer yield, process rate, and catalyst stability improvement compared to Phase 1 for multiple types of whole biomass (four or more sources, e.g., corn stover, palm frond, coconut husk). <u>Deliverable:</u> Report summarizing milestone data. 23 Months – <u>Milestone:</u> Demonstrate delignification, monomer yield, process rate, and stability improvement consistent with End of Program Metrics. <u>Deliverable:</u> Report summarizing milestone data. <u>Deliverable:</u> Refined TEA and process model for T&E evaluation. End of Program Milestone – 24 Months: <u>Milestone:</u> Achieve $\geq 90\%$ delignification and 100% monomer yield with maximized Space Time Yield; T&E will evaluate chemical

	value with comprehensive TEA. <i>Deliverable:</i> End of Phase report summarizing progress against all tasks, metrics, and milestones.
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Table 3: Anticipated milestones and deliverables for all FAs.

Preliminary System Design Report shall include an initial process model of the chosen process unit operations and catalyst system in ASPEN Plus or similar chemical engineering process modeling software. This model will be the basis and should be incorporated into a full process TEA at multiple scales. Performers should explore a range of scales, at least 3, spanning small, modular production in a deployed scenario to large chemical manufacturing.

For the Catalyst Development Metric and associated milestones and deliverables, proposers must define initial catalyst stability, activity, and selectivity targets for improvement. For each end of phase and end of program milestones, performers should aim to maximize space time yield for their chosen catalyst system and process unit operations across three production scales, enabling scalability and enabling commercial application.

During the program, performers will be required to provide sufficient information to the Government T&E Team to allow for additional testing and evaluation which may include catalyst synthesis parameters, reaction parameters, and chemical characterization data. In addition, small volumes of catalyst or chemical samples may be requested by the Government T&E Team for testing and evaluation.

1.7 Test and Evaluation (T&E)

Additional testing and evaluation support will be provided by the Government T&E Team during Phases 1 and 2. The Government T&E Team will consist of national laboratories with expertise in biomass processing and conversion, for example, National Laboratory of the Rockies (NLR) and Idaho National Laboratory (INL). The government T&E teams are not eligible to be a part of any competitive proposer team. Government T&E teams cannot answer questions about this solicitation; all questions related to this solicitation must be sent to Fleetwood@darpa.mil. They will provide the following types of support to all selected performer teams:

- Government furnished sourcing and preprocessing of whole biomass
- Government furnished sourcing of scaled process intermediates, such as minimally process biomass feedstocks, separated lignin oil, and cellulose/hemicellulose/lignin blendstocks

The Government T&E Team is available to provide the following types of support to all selected performer teams:

- Analytical support of government furnished biomass to include biomass characterization and compositional analysis
- Support to review and assist in the development of rigorous TEA and process models
- Product chemical evaluation and testing to include analytical and standardized testing
- Note: While performers may leverage the capabilities listed above, they may propose similar capabilities that meet or exceed the Fleetwood objectives.

Performers must provide TEA and process models for their approach, across three production scales, to the DARPA Program Manager and the Government T&E Team for assessment. In

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addition, performers are responsible for developing an analytical test and evaluation strategy for iterative catalyst and process development, to be refined with support from the Government T&E Team. This testing is anticipated to include a combination of analytical techniques and must be sufficient to demonstrate progress against program metrics. A number of key program milestones rely on effective execution of this testing strategy over the life of the program.

Performers should plan to interact with the Government T&E team frequently over the course of the program. This team will be composed of researchers with relevant expertise from US Government laboratories. Performers will engage with T&E team to consult on experimental design and relevant controls, in addition to TEA assumptions, calculations, and reporting. The interaction with T&E team, including early consideration of TEA and experimental results, is envisioned to de-risk and inform eventual transition pathways. Additionally, the Government T&E team will replicate and evaluate key aspects of successful technologies, as possible, at laboratory scale. Consequently, proposals must plan for any required agreements needed to share background intellectual property (IP) with government partners that could be required for evaluation studies. These agreements should be appropriate for the proposed technology and experimental studies, and they may necessitate material transfer agreements, data sharing agreements, cooperative research and development agreements (CRADAs), or other such agreements. DARPA is not responsible for negotiating these agreements because these terms must be agreed upon between the two parties exchanging information and/or materials (i.e., the government partner and the contracted performer on the program). Any foreground IP generated during performance on Fleetwood will be governed by the policy set forth in Section 5.5 and the terms and conditions in the model Other Transaction for Prototype Agreement), and as such, this information will be shared with members of the government team, as necessary, for them to support the program.

1.8 Program Security

At this time, the Government expects much of the work performed under Fleetwood to be unclassified, fundamental research but potentially still subject to pre-publication review. Information generated that does not clearly identify as “CUI” (controlled unclassified information) may still need to undergo review prior to public release. If abstracts or OPPs include novel processes or steps to manufacture chemical compounds listed on the commerce control list, explosives listed in the ANNEX to Category 1 List of Explosives, or the US Munitions List proposers are encouraged to first contact the agency point of contact listed in the Overview section prior to the abstract due date to discuss security requirements. Press releases and other marketing and publicity materials are not considered to be results of research and therefore are always subject to the public release process. All other publications, articles, and scientific presentations must be submitted to the Government for review and approval 30 days in advance of the required submission date, to allow for the removal of any sensitive information. It is anticipated that milestone reviews will be used to work toward mutually agreeable plans for reviewing publications, methods, and data, prior to release, involving Government partners, DARPA, and performers.

2. PROGRAM SOLICITATION AUTHORITY

This PS may result in the award of an Other Transaction (OT) for Prototype agreement, which can include not only commercially-available technologies fueled by commercial or strategic

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investment, but also concept demonstrations, pilots, and agile development activities that can incrementally improve commercial technologies, existing Government-owned capabilities, and/or concepts for broad defense and/or public application(s). The Government reserves the right to award an OT for Prototype under 10 U.S.C. § 4022 or make no award at all. In all cases, the Government agreements officer shall have sole discretion to negotiate all agreement terms and conditions with selected proposers. The OT agreement will not require cost sharing unless the proposer is a traditional defense contractor who is not working with a non-traditional defense contractor participating in the program to a significant extent.

2.1 Program Solicitation Procedure

Proposers responding to this solicitation must submit a 5-page abstract as described in Section 4.2. This process enables the Government to evaluate:

- (1) Whether the proposers understand the key challenges of the Fleetwood program and offers a compelling and technically feasible approach to solve those challenges.
- (2) Whether proposers are capable of executing the proposed concept.

Abstracts will be reviewed based on the evaluation criteria outlined in Section 4.3. Proposers whose abstracts meet these criteria may be invited to submit an Oral Presentation Package (OPP) and participate in an oral presentation, as detailed in Section 4.4. Final evaluation criteria for oral presentations are provided in Section 4.5. The Government will determine which proposers may be awarded an OT for Prototype agreement based on these evaluations. NOTE: The Government will not reimburse proposers for costs associated with abstract submissions or oral presentations.

Fleetwood Source Selection Process:

- a. **Industry Day (Optional):** DARPA will host an Industry Day on March 12, 2026, where the Program Manager will present the program's goals and address questions from the participants. A comprehensive questions and answers (Q&A) document summarizing the session will be posted online after the event. Participation in Industry Day is optional and is not required for submitting an abstract. Additional details will be provided in Special Notice DARPA-SN-26-49 separate from this PS.
- b. **Program Solicitation Questions and Answers (Q&A) (Informational Only):** DARPA will consolidate and post a Q&A document from the Industry Day online at the Fleetwood program page (<https://www.darpa.mil/research/programs/fleetwood>). Questions submitted after Industry Day can be emailed to Fleetwood@darpa.mil. Relevant clarifications will be addressed and posted prior to the final abstract submission deadline at the Fleetwood program page (<https://www.darpa.mil/research/programs/fleetwood>).
- c. **Abstracts (Required):** Abstracts must be submitted as specified in Section 4.2. This will allow the Government to assess proposer's technical comprehension and understanding of the key challenges of the Fleetwood program and ability to execute the proposed concept (see Section 4.3). Only proposers who submit an abstract will be considered for further participation in the Fleetwood program. Selected proposers will be invited to submit an OPP and deliver an in-person oral presentation (see Section 4.4).
- d. **Oral Proposal Package (OPP)/Oral Presentation (Required if selected):** Selected proposers will submit an OPP and participate in an in-person oral presentation

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approximately two weeks after notification. Detailed instructions and format are contained in Section 4.4. Final requirements, templates, submittal instructions, and presentation dates will be provided in the invitation to submit an OPP. OPPs will be reviewed based on criteria outlined in Section 4.5. Reviews will not be made public or provided to other proposers. For Phase 1 and 2, proposers must propose an OT for Prototype with fixed payable milestones. Milestones represent completed events tied to observable technical achievements in the path to accomplish program objectives, with payments triggered upon successful completion. A Schedule of Milestones and Payments is included in the Model Other Transaction for Prototype Agreement.

2.2 Program Length

The Government will review OPPs and oral presentations to determine which proposed solutions sufficiently meet the evaluation criteria stated in Section 4.5. Upon favorable review, and subject to the availability of funds, the Government may award an OT for Prototype under 10 U.S.C. § 4022 with fixed, payable milestones. Phase 1 (base period) will be 12 months, and Phase 2 (option period) will be 12 months. **This PS is requesting a proposal for both phases and to address both FAs.** The cost proposal should include cost breakouts for both phases and with the same level of details and support documentation. It is noted that the Government is not required to exercise the option period at all, which is subject to the availability of funds and is dependent on the program goals during and after Phase 1.

3. ELIGIBILITY INFORMATION

3.1 Eligible Applicants

3.1.1 Federally Funded Research and Development Centers (FFRDCs) and Government Entities

Federally Funded Research and Development Centers (FFRDC), University Affiliated Research Centers (UARCs), and Government Entities to include National Laboratories are not eligible to propose to this solicitation.

3.1.2 Other Applicants

Non-U.S. organizations and/or individuals may participate to the extent that such participants comply with any necessary nondisclosure agreements, security regulations, export control laws, and other governing statutes applicable under the circumstances.

3.2 Organizational Conflicts of Interest (OCI)

Without prior approval or a waiver from the DARPA Deputy Director, a contractor cannot simultaneously provide scientific, engineering, technical assistance (SETA), advisory and assistance services (A&AS), or similar support and also be a technical performer. As part of the OPP, all members of the proposed team (including any potential subawardees or consultants) must affirm whether they (their organizations and individual team members) are providing SETA or similar support to any DARPA office(s) through an active award or subaward. All facts relevant to the existence or potential existence of Organizational Conflicts of Interest (OCI) must be disclosed in the Administrative and National Policy Requirements document, should the proposer be invited to submit an OPP.

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If SETA, A&AS, or similar support is being or was provided to any DARPA office(s), the OPP must include in the Administrative and National Policy Requirements document:

- The name of the DARPA office receiving the support;
- The prime contract number;
- Identification of proposed team member (subawardee, consultant) providing the support;
- An OCI mitigation plan.

Under this section of the OPP, the proposer is responsible for providing this disclosure with each OPP submitted. The disclosure must include the proposer's OCI status, and as applicable, proposed team member's OCI mitigation plan. The OCI mitigation plan must include a description of the actions the proposer has taken, or intends to take, to avoid, neutralize, or mitigate such conflict, prevent the existence of conflicting roles that might bias the proposer's judgment, and prevent the proposer from having unfair competitive advantage. Prior to the start of OPP evaluations, the Government will assess potential conflicts of interest based on the OPPs submitted. The Government will promptly notify the proposer if any appear to exist. The Government assessment does NOT affect, offset, or mitigate the proposer's responsibility to give full notice and planned mitigation for all potential organizational conflicts.

If, in the sole opinion of the Government after full consideration of the circumstances, a proposal fails to fully disclose potential conflicts of interest and/or any identified conflict situation cannot be effectively mitigated, the OPP will be rejected without technical evaluation and withdrawn from further consideration for award.

If a prospective proposer believes a conflict of interest exists or may exist (whether organizational or otherwise) or has questions on what constitutes a conflict of interest, the proposer should send his/her contact information and a summary of the potential conflict via the specific email address identified in this PS before time and effort are expended in preparing an OPP and mitigation plan.

4. GUIDELINES FOR ABSTRACTS AND ORAL PRESENTATION PACKAGES

4.1 General Guidelines

- a. Do not include elaborate brochures or marketing materials; only include information relevant to the submission requirements or evaluation criteria.
- b. Use of a diagram(s) or figure(s) to depict the essence of the proposed solution is encouraged.
- c. All Abstracts, Oral Presentations, and Proposals shall be unclassified.
- d. Proposers are responsible for clearly identifying proprietary information. Submissions containing proprietary information must have the cover page and each [age containing such information clearly marked with a label such as "Proprietary" or "Company Proprietary." NOTE: "Confidential" is a classification marking used to control the dissemination of U.S. Government National Security Information as dictated in Executive Order 13526 and should not be used to identify proprietary business information.

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- e. Questions can be sent to Fleetwood@darpa.mil by March 25, 2026 5:00 PM (ET).
- f. All abstracts are required to be submitted via DARPA's Broad Agency Announcement Tool (BAAT) by **TBD**. Files containing Controlled Unclassified Information (CUI) must be appropriately marked and must be encrypted when sending over the Internet. Please visit [Proposer Instructions and General Terms and Conditions](#) for instructions on how to submit your abstract through DARPA's BAAT. It is important to note that the terms and conditions on the remainder of the Proposer Instructions and General Terms and Conditions link above do not apply to this solicitation. The purpose of referencing the website is for you to obtain instructions for DARPA's BAAT.
- g. Submissions sent through other mediums, channels, or after the prescribed PS deadline will not be considered, reviewed, nor evaluated.
- h. Proposers providing abstracts that are not invited to an Oral Presentation will be notified in writing as soon as practicable.

4.2 Abstract Content

Abstracts shall not exceed five (5) single-sided 8.5" by 11" written pages using 12-point Times New Roman font with 1" margins all around.

Abstracts must include the following:

1. **Abstract Summary Slide:** See Attachment A: Abstract Summary Slide Template
2. **Cover Sheet:** See Attachment B. Must include Proposer Name, Title, and Name, Email Address, Phone, Address for Technical and Administrative Points of Contact. (The Cover Sheet does not count against page limits).
 - The proposer shall include a statement that no people on the proposer's teamwork for DARPA as Scientific Engineering Technical Assistance (SETA), Advisory and Assistance Services (A&AS) or similar support services, as DARPA has a policy prohibiting such people from working as a technical performer. Include this statement on the title page; it will NOT count as part of the five (5) written pages limit.
3. **Executive Summary:** Provide a summary of the technical goals and challenges of Fleetwood. This summary shall be stated in the proposer's own words without any "copy and paste" of this solicitation. The goal is for the proposer to demonstrate clear understanding of Fleetwood's purpose and goals as well as what they think the primary risks are to successful development of the technology. The summary shall be no more than 1 page and is included in the five (5) written pages limit.
4. **Proposed Technical Approach:** Provide a summary of the technical approach to address the goals and challenges stated above. The goal is for the proposer to demonstrate a clear and innovative approach to achieve the program metrics. The summary shall be no more than 2 pages and is included in the five (5) written pages limit.
5. **Technical Challenges:** Identify specific technical challenges associated with the proposed approach. The proposer should include what they think the primary risks are to successful development in the Fleetwood program and the envisioned mitigations for those risks. The summary shall be no more than 1 page and is included in the five (5) written pages limit.

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6. **Technical Ability:** Detail why the proposer's team believe that they have the ability to be successful at achieving the goals, if selected, for Fleetwood. The proposer may include past experience, organizational capabilities, team members' qualifications, or anything else that demonstrates competence in designing and executing Fleetwood's goals. The summary shall be no more than 1 page and is included in the five (5) written pages limit.
7. **Estimated Cost:** Include the estimated total labor cost, and estimated materials and ODCs (equipment, materials, travel, tuition) for Phase 1 and 2. This may be presented as no more than half a page narrative or table and is included in the five (5) written page limit.
8. **Bibliography:** Provide a list of citations, references, or end notes. References are not included in the page limit.

4.3 Abstracts – Process and Basis of Evaluation

Abstract evaluation criteria are listed in descending order of importance. Individual abstracts will be evaluated against the evaluation criteria described below:

- a. **Technical Approach:** The proposed technical approach is reasonable, feasible, and innovative. The approach demonstrates an innovative yet feasible approach to address the identified technical risks and challenges and meet program metrics.
- b. **Technical Ability:** The proposers demonstrate an ability, if selected, to achieve the goals of the Fleetwood program.
- c. **Technical Comprehension:** The technical understanding is accurate, and key technical challenges and risks are identified.

Abstracts will be evaluated by the Government using the evaluation criteria listed above. The Government will use the evaluation criteria to assess similarities, differences, strengths, and weaknesses of the competing abstracts and, ultimately, use that assessment to determine the selection of those proposers offered the opportunity to proceed to Oral Presentations. The Government will endeavor to complete the evaluation of abstracts within 10 business days of the closing of the submittal period. As stated above, proposers are required to submit an abstract for evaluation by the Government to minimize effort and reduce the potential expense of preparing an unsuccessful proposal. The Government will respond to the 5-page abstract with a statement as to whether the Government is interested in seeing a 30 minute (approximately 20 minutes presentation, 10 minutes question and answer period) Oral Presentation. If the Government is not interested in an Oral Presentation, it will state this in an email to the proposer. Upon review of abstracts, the Government may elect to invite all, some, or none of the proposers to Oral Presentations. *Only abstract proposers invited by the Government to participate in Oral Presentations are eligible to provide one.*

4.4 Oral Proposal Package (OPP)/Oral Presentation Content

If the Government expresses interest in an Oral Presentation, the proposer will be asked to provide a presentation to provide further details on its proposed solution. Specific instructions (including content submission guidelines and relevant templates) will be provided in the invitation to participate. If selected, proposers can expect to be asked to provide the following information:

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- **Title Page** – Proposer Name, Title, Date, Point of Contact (POC) Name, Email Address, Phone, and Address. (The title page does not need to be briefed, and it may be included as the first slide in the deck for the oral presentation). The title page must also include the following:
 - A statement that no people on the proposer’s team works for DARPA as a SETA, A&AS, or similar support services on an active contract or subcontract (including those awarded through DARPA agents); or list which offices the proposer supports and identify the prime contract numbers. DARPA policy prohibits supporting contractor individuals and entities from concurrently working as research and development performers, unless potential organizational conflicts of interest are identified, eliminated, or appropriately mitigated, and granted a waiver.
 - A statement that identifies and substantiates which of the following condition(s) are met to permit use of OTs for Prototype in accordance with 10 U.S.C. § 4022(d)(1): (A) There is at least one nontraditional defense contractor or nonprofit research institution participating to a significant extent in the prototype project; (B) All significant participants in the transaction other than the Federal Government are small businesses (15 U.S.C. § 638) or nontraditional defense contractors; (C) At least one third of the total cost of the prototype project is to be paid out of funds provided by sources other than the Federal Government; or (D) The senior procurement executive for the agency determines in writing that exceptional circumstances justify the use of a transaction that provides for innovative business arrangements or structures that would not be feasible or appropriate under a contract, or would provide an opportunity to expand the defense supply base in a manner that would not be practical or feasible under a contract.
- **Task Description Document (TDD)** – will be provided to the proposers at a later date.
- **Detailed Cost Proposal Spreadsheet** – Complete cost information must be provided for Phase 1 and Phase 2.
- **Schedule of Milestones and Payments.**
- **Completed Representations and Certifications** – will be provided to the proposers at a later date.
- **Oral Proposal Package Slides** – Refer to the information the invitation letter. Please note, these oral presentations will be in-person only in the Washington, DC metro area at a location to be determined. No virtual presentations will be allowed.

In addition to the above required areas, the Government may request the proposer to provide additional information. Proposers should expect to have approximately 20 minutes for presentation and approximately 10 minutes to address any questions from the Government panel. Oral presentations are subject to the following constraints:

- No more than 10 slides in PDF or PowerPoint format
- No smaller than 10-point font
- Video demonstrations are allowed to be embedded in the slide deck

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- All presented material is to be submitted to BAAT by **TBD**. Files containing Controlled Unclassified Information (CUI) must be appropriately marked and must be encrypted when sending over the Internet.

4.5 Oral Presentations – Process and Basis of Evaluation

Oral presentation evaluation criteria will be provided with the invitation to present. The Government intends for Oral Presentations to be done in-person over the course of 1-2 days in the Washington, DC metro area; the Government reserves the right to record presentations. The Government will evaluate information provided in the content submission (documentation), the Oral Presentation, and Q&A session as basis for evaluation. Oral Presentations will be evaluated by the Fleetwood Program Manager with support from a panel composed of Government subject matter experts (SMEs).

5. AWARDS

5.1 General Guidelines

Upon favorable review of the proposal and subject to the availability of funds, the Government may choose to award an OT for Prototype agreement for Phase 1 and Phase 2.

The Agreements Officer reserves the right to negotiate directly with the proposer on the terms and conditions prior to execution of the resulting OT agreement, including payment terms, and will execute the agreement on behalf of the Government. A copy of the draft OT agreement is attached to this PS for review. In order to speed up negotiations, proposers selected for oral presentations will be required to either attest to compliance of all OT agreement articles or note those they take exception to. Be advised, only a Government Agreements Officer has the authority to enter into, or modify, a binding agreement on behalf of the United States Government.

In order to receive an award:

- a. Proposers must have a Unique Identity ID number and must register in the System for Award Management (SAM). Proposers are advised to commence SAM registration upon notification of entry to Phase 1 of the competition.
- b. Proposers must also register in the prescribed Government invoicing system (Wide Area Work Flow: <https://wawf.eb.mil/xhtml/unauth/registration/notice.xhtml>). The DARPA Invoices team will provide assistance to those proposers from whom a proposal is requested.
- c. Proposers must be determined to be responsible by the Agreements Officer and must not be suspended or debarred from award by the Federal Government nor be prohibited by Presidential Executive Order and/or law from receiving an award.
- d. Being asked to submit a proposal does not guarantee that a proposer will receive an award. The Government reserves the right not to make an award.

5.2 Controlled Unclassified Information (CUI) and Controlled Technical Information (CTI) on Non-DoD Information Systems

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Further information on Controlled Unclassified Information identification, marking, protecting and control, to include processing on Non-DoD Information Systems, is incorporated herein and can be found at <http://www.darpa.mil/work-with-us/additional-baa>. A program-specific CUI Guide has not been established to help proposers determine CUI thresholds for information relevant to, and technologies developed under the program at this time. CTI is not anticipated for this program but, foreign proposers are encouraged to understand U.S. export law and have a plan in place to obtain export licenses if proposing to generate export-controlled information. Possible methods include teaming with a U.S. prime and/or having a U.S. subsidiary/parent company. Dependent upon selection for award, a program specific CUI guide will be provided to the performer to observe and follow if export-controlled information will be generated. If abstracts or OPPs include novel processes or steps to manufacture chemical compounds list on the commerce control list, explosives listed in the ANNEX to Category 1 List of Explosives, or the US Munitions List proposers are encouraged to first contact the agency point of contact listed in the Overview section prior to the abstract due date to discuss security requirements.

5.3 Representations and Certifications

All proposers are required to submit DARPA-specific representations and certifications for Prototype OT awards in order to be eligible to receive an OT award. See <https://www.darpa.mil/research/opportunities/rebs-certs> for further information on required representations and certifications for Prototype OT awards.

5.4 Competition Sensitive Information

DARPA policy is to treat all submissions as competition sensitive, and to disclose their contents only for the purpose of evaluation. Restrictive notices notwithstanding, during the evaluation process, submissions may be handled by support contractors for administrative purposes and/or to assist with technical evaluation. All DARPA support contractors performing this role are expressly prohibited from performing DARPA sponsored technical research and are bound by appropriate nondisclosure agreements. Input on technical aspects of the proposals may be solicited by DARPA from non-Government consultants/experts who are strictly bound by the appropriate non-disclosure requirements.

5.5 Intellectual Property (IP) / Data Rights

The Government requires, at a minimum, Government Purpose Rights for the technology developed under the program. IP proposals should, at a minimum, allow DARPA to:

- Flexibly brief U.S. Government stakeholders regarding technical progress and accomplishments,
- Allow validation of technical performance, capabilities, and accomplishments by independent technical (potentially non-Government) experts, subject to NDA restrictions,
- Facilitate discussion of technical challenges and applications with the broader technical community – for example, by starting a new DARPA program that attempts to solve a serious technical challenge that limits further progress,
- Support of analyses of alternatives, and

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- Support transition opportunities, including design and performance data required to support other acquisition activities. These latter activities may require the Government to conduct an independent performance analysis.

Proposers responding to this PS shall appropriately identify any potential restrictions on the Government's use of an IP furnished by the proposer. This includes both Noncommercial Items and Commercial Items. Proposers are encouraged to identify these restrictions in a format like the table depicted below. If no restrictions are intended, then the proposer should state "NONE."

List of restrictions

Technical Data, Computer Software to be Furnished with Restriction	Summary of Intended Use in the Conduct of the Research	Basis for Assertion	Asserted Rights Category	Name of Person Asserting Restrictions
(LIST)	(NARRATIVE)	(LIST)	(LIST)	(LIST)

5.6 Procurement Integrity Act (PIA)

All awards under this PS shall be treated as Federal Agency procurements for purposes of 41 U.S.C. Chapter 21. Accordingly, the PS competitive solicitation process and awards made thereof must adhere to the ethical standards required by the PIA.

5.7 Fundamental Research Risk-Based Security Review Program (FRRBS)

DARPA's Fundamental Research Risk-Based Security Review Process (formerly CFIP, now FRR-BS a.k.a. "FRRBS") is an adaptive risk management security program designed to help protect the critical technology and performer intellectual property associated with DARPA's research projects by identifying the possible vectors of undue foreign influence. DARPA will create risk assessments of all proposed senior/key personnel selected for negotiation of a fundamental research award. The DARPA risk assessment process will be conducted separately from the DARPA scientific review process and adjudicated prior to final award. For additional information on this process, please visit [Proposer Instructions: Other Transactions](https://www.darpa.mil/about/offices/contracts-management/proposer-transactions) (<https://www.darpa.mil/about/offices/contracts-management/proposer-transactions>).

All submissions proposing to this solicitation must complete the Common Disclosure Forms, biosketch, and current/pending support forms. Forms can be found here:

- [Common Form for Biographical Sketch \(nsf.gov\)](https://www.nsf.gov/pubs/2014/pub14101)
- [Common Form for Current and Pending \(Other\) Support \(nsf.gov\)](https://www.nsf.gov/pubs/2014/pub14102)

6. PS DEFINITIONS

"Data" refers to recorded information, regardless of form or method of recording, which includes but is not limited to, technical data, software, mask works and trade secrets. The term does not include financial, administrative, cost, pricing or management information and does not include inventions.

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“Government Purpose” means any activity in which the United States Government is a party, including cooperative agreements with international or multi-national defense organizations, or sales or transfers by the United States Government to foreign governments or international organizations. Government purposes do not include the rights to use, modify, reproduce, release, perform, display, or disclose technical data for commercial purposes or authorize others to do so.

“Government Purpose Rights” are rights to use, duplicate, release, or disclose, Data, in whole or in part, in any manner, for Government purposes only, and to have or permit others to do so for Government purposes only.

“Nontraditional Defense Contractor” is defined in 10 U.S.C. § 3014 as an entity that is not currently performing and has not performed, for at least the one-year period preceding the solicitation of sources by the DoD for the procurement or transaction, any contract or subcontract for the DoD that is subject to full coverage under the cost accounting standards prescribed pursuant to 41 U.S.C. § 1502 and the regulations implementing such section. This includes all small business concerns under the criteria and size standards in 15 U.S.C. § 632 and 13 C.F.R. Part 121.

“Other Transaction” refers to the type of OT that may be awarded as a result of this PS. This type of OT is authorized by 10 U.S.C. § 4022 for prototype projects directly relevant to enhancing the mission effectiveness of military personnel and the supporting platforms, systems, components, or materials proposed to be acquired or developed by the DoD, or for the improvement of platforms, systems, components, or materials in use by the armed forces.

“Prototype Project” is described in the DoD Other Transactions Guide (Version 1, Nov. 2018) issued by the Office of the Under Secretary of Defense for Acquisition and Sustainment: <https://www.acq.osd.mil/asda/dpc/cp/policy/docs/guidebook/Guide%20to%20Research%20Other%20Transactions.pdf>

“Small Business Concerns” is defined in the Small Business Act (15 U.S.C. § 632).